Midhun Somanunnithan

Machine Learning & Computer Vision | LLM Systems | PyTorch | Python

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Core Skills: PyTorch · computer vision · deep learning · DINOv2 · Vision Transformers · metric learning · image retrieval · FAISS · OpenCV · image segmentation · LLM systems · agentic AI · tool-executing agents · language modeling · held-out evaluation · Python · Git · Linux · SLURM · A100 GPUs

PROFILE

Deep Learning Research Assistant and E-Mobility (ACES) master's student at FAU Erlangen-Nürnberg. I build and evaluate computer-vision and machine-learning systems in PyTorch - my best image-retrieval pipeline reached 83.4% mAP and 100% Recall@10 across seven benchmarked designs - and I have shipped JARVIS, an agentic voice assistant with provider fallback, tool execution, and persistent memory. I also built neural networks and language models from scratch. I value reproducible experiments, honest evaluation, and connecting models to real software and physical processes. English C1, German B2. Available immediately.

EXPERIENCE

Deep Learning Research Assistant - Computer Vision
FAU Erlangen-Nürnberg | February 2026 - Present

- Owned seven image-retrieval benchmark experiments end to end in PyTorch - design, training, and evaluation - best at 83.4% mAP and 100% Recall@10.
- Compared metric-learning objectives (Triplet, ProxyAnchor, ArcFace) and CNN versus Vision Transformer backbones, with SAM crops and YOLOv11 pose features.
- Ran ablations and systematic failure analysis to separate real improvements from noise, reproducible on A100 GPUs with Linux, Git, and SLURM.

Research Assistant - Automation and Mechatronics
FAU Institute of Manufacturing Technology | 2026 - Present

- Built an automated spray-booth workflow for repeatable sample preparation with multi-axis motion and pneumatic control.
- Planned parameter studies and validated repeatability on real physical equipment, and documented the setup for reuse.

Data Science and Machine Learning Intern
Entri Elevate | October 2023 - December 2023

- Cleaned and prepared tabular data and engineered features in Python with Pandas and NumPy.
- Trained models with cross-validation and evaluated results.

SELECTED PROJECTS

Fine-Grained Image Retrieval Benchmark (Vase CBIR) Content-based image retrieval on historical vase illustrations. Built seven controlled pipelines in PyTorch - a ResNet50 baseline, then Triplet, ProxyAnchor, and ArcFace metric-learning heads, SAM-cropped variants, a YOLOv11 figure-pose model, and a best DINOv2 ViT-S/14 model reaching 83.4% mAP and 100% Recall@10 (FAISS). Shared extract/train/retrieve/evaluate structure, folder-level split, and robustness and failure analysis across viewpoint, occlusion, and look-alike objects.

JARVIS - Agentic Voice AI Assistant Real-time Windows voice assistant (React and FastAPI, WebSocket streaming) with an LLM provider chain and automatic fallback, an inspectable action-tag tool protocol (web search, file I/O, shell), Gmail and Calendar through OAuth2, screen vision, persistent learned memory, and confirmation gates for sensitive actions.

Deep Learning from Scratch Neural networks and language models built by hand in PyTorch, from a scalar autograd engine and a bigram model to a 6-layer BatchNorm MLP (47,024 parameters) on 32,033 names, improving held-out negative log-likelihood from 2.45 to 2.09. Diagnosed training with activation and gradient histograms and the update-to-data ratio, using Kaiming initialization and BatchNorm - a Transformer stage is next.

AI-Assisted Automated Spray Process Projektarbeit at FAU LFT. Built an automated spray system (GRBL motion, pneumatic actuation) and a Python/OpenCV pipeline that quantifies spray-pattern quality. Trained CNN and Vision Transformer models in PyTorch to grade quality, explored U-Net/YOLO segmentation for defect detection, built a multimodal model from images and process parameters, and used Bayesian optimization to select parameters.

EDUCATION

M.Sc. E-Mobility (ACES)
Friedrich-Alexander-Universität Erlangen-Nürnberg | 2022 - Present

B.Tech Mechanical Engineering
APJ Abdul Kalam Technological University | 2017 - 2021

TECHNICAL SKILLS

Computer Vision: OpenCV, DINOv2, Vision Transformers, YOLOv11 (incl. pose), ResNet50, SAM, image retrieval (FAISS), metric learning, image segmentation (U-Net/YOLO), robustness and failure analysis

Deep Learning: PyTorch, neural networks from scratch, autograd, BatchNorm, language modeling, TensorFlow/Keras, scikit-learn

LLM and Systems: LLM API integration, model routing and fallback, tool-executing agents, FastAPI, React, WebSockets, OAuth2

Programming and Tools: Python, Git, Linux, SLURM, A100 GPUs, basic C/C++

Data and Methods: NumPy, Pandas, feature engineering, multimodal modeling, Bayesian optimization, benchmarking, ablations, held-out evaluation, failure analysis

ADDITIONAL INFORMATION

Open-source work: GitHub: JARVIS voice agent, Deep Learning from Scratch, vase retrieval benchmark

Availability: Immediately, or by agreement

Location: Nuremberg / Erlangen, Germany (open to relocation and remote)

LANGUAGES

Languages: English - C1 • German - B2, working proficiency